NEET UG PHYSICS • STANDARD ALIGNMENT

Thermodynamics

Syllabus Scope: Thermal equilibrium and Zeroth law; Heat, internal energy, and work; First law of thermodynamics; Specific heat capacities ($C_p - C_v = R$); State variables and equations of state; Quasi-static, isothermal, adiabatic, isochoric, isobaric, and cyclic processes; Second law of thermodynamics; Reversible and irreversible processes; Carnot engine and Carnot's theorem.

1. Learning Objectives & Chapter Overview

Thermodynamics is the macroscopic science dealing with the concepts of heat and temperature and the inter-conversion of heat and other forms of energy. Unlike mechanics (which studies the motion of particles or bodies as a whole under forces and torques) or kinetic theory (which describes molecular velocity distributions), thermodynamics describes bulk systems using directly measurable macroscopic variables such as pressure (P), volume (V), temperature (T), mass (m), and composition.

Key Competencies for NEET UG:

- Distinguish between microscopic/mechanical states and macroscopic thermodynamic equilibrium states.
- Apply the Zeroth Law to establish the concept of temperature and thermal equilibrium across diathermic and adiabatic boundaries.
- Clearly differentiate between state variables (like internal energy U) and path-dependent energy transfer modes (heat Q and work W).
- Solve quantitative problems using the First Law of Thermodynamics ($\Delta Q = \Delta U + \Delta W$) across isothermal, adiabatic, isobaric, isochoric, and cyclic processes, including P-V indicator diagrams.
- Evaluate the performance limits of heat engines and refrigerators using the Second Law of Thermodynamics and Carnot cycle relations.

2. Core Concepts & Detailed Explanations

2.1 Historical Context: From Caloric Fluid to Energy

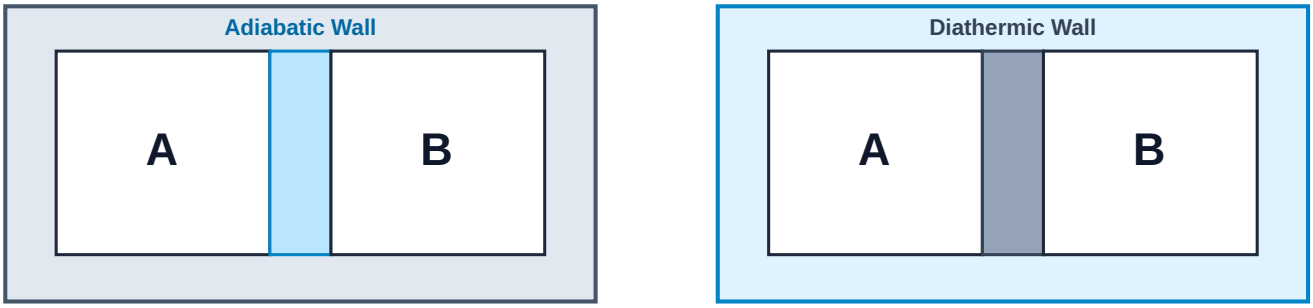
Historically, heat was imagined as an invisible weightless fluid called *caloric* filling the pores of substances, flowing from hotter to colder bodies much like water equalizing levels between connected tanks. This caloric model was overturned in 1798 by Benjamin Thomson (Count Rumford). While observing the boring of a brass cannon, Rumford noted that the large amount of heat generated (enough to boil water) depended directly on the mechanical work done by the horses turning the drill, and *not* on the sharpness of the drill. If caloric fluid were being scooped out of pores, a sharper drill would release more heat; instead, dull drills still generated continuous heat as long as mechanical work was supplied. This established that heat is a form of energy and that mechanical work can be converted into heat (e.g., rubbing palms together in winter) and heat into mechanical work (e.g., a steam engine moving pistons).

Mechanics vs. Thermodynamics: When a bullet is fired from a gun, its mechanical state (bulk kinetic energy) changes, not its temperature. Only when the bullet pierces wood and stops does its bulk kinetic energy convert into internal disordered energy, raising the temperature of the bullet and the wood.

2.2 Thermal Equilibrium: Adiabatic and Diathermic Walls

In mechanics, equilibrium means the net external force and net external torque on a system are zero. In thermodynamics, a system is in an **equilibrium state** if the macroscopic variables characterizing the system (P, V, T, m, composition) do not change with time.

- **Adiabatic Wall:** An insulating wall (which may be movable) that does *not* allow the flow of heat energy from one system to another. If two gases A and B with variables (P_A, V_A) and (P_B, V_B) are separated by an adiabatic wall and insulated from the surroundings, any possible pair of values (P_A, V_A) is in equilibrium with any possible pair (P_B, V_B).
- **Diathermic Wall:** A conducting wall that allows heat energy to flow from one system to another. When an adiabatic wall between gases A and B is replaced by a diathermic wall, their macroscopic variables change spontaneously to new values (P'_A, V'_A) and (P'_B, V'_B) until energy flow ceases. At this point, A and B are in **thermal equilibrium** with each other, and their temperatures are equal. (If the containers have fixed volumes, only pressures change to reach equilibrium.)



(a) No heat flow between A and B **(b) Heat flows until thermal equilibrium**

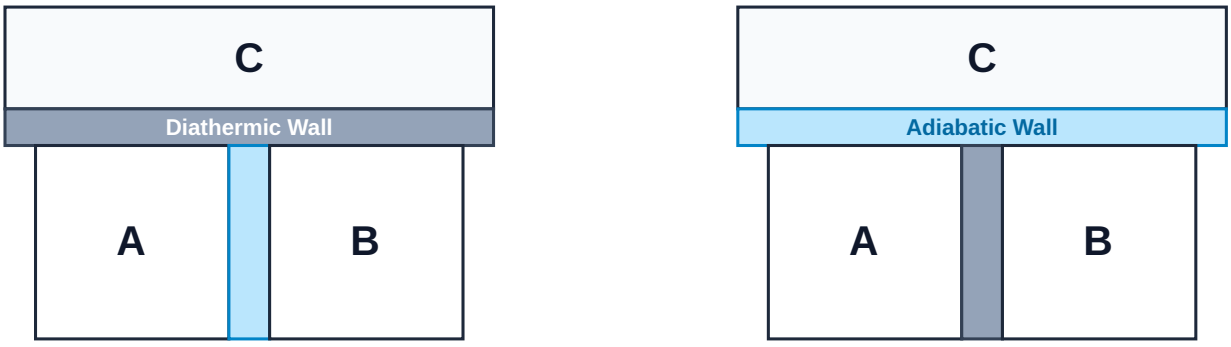
Tutorial Diagram 1: (a) Systems A and B separated by an insulating adiabatic wall. (b) Systems A and B separated by a conducting diathermic wall, attaining thermal equilibrium.

2.3 Zeroth Law of Thermodynamics

Formulated by R.H. Fowler in 1931, the **Zeroth Law of Thermodynamics** states:

> **Zeroth Law of Thermodynamics:** "Two systems in thermal equilibrium with a third system separately are in thermal equilibrium with each other."

Suppose systems A and B are separated from each other by an adiabatic wall, while both are in contact with a third system C via a conducting (diathermic) wall. Once A and B separately reach thermal equilibrium with C, if the wall between A and B is made diathermic and C is insulated by an adiabatic wall, no further change occurs in the states of A and B. This establishes the existence of a scalar thermodynamic state variable called **temperature** (T): if $T_A = T_C$ and $T_B = T_C$, then $T_A = T_B$.



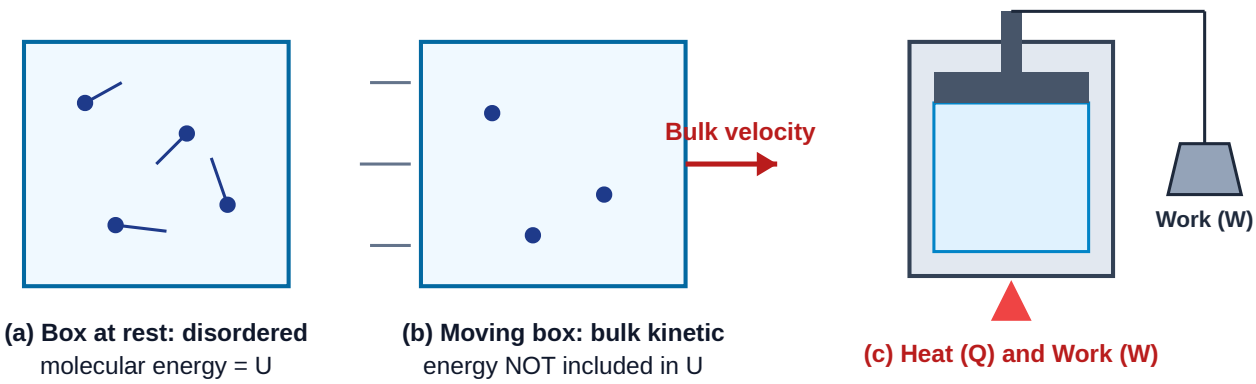
(a) A and B in equilibrium with C **(b) A and B found in equilibrium**

Tutorial Diagram 2: Experimental verification of the Zeroth Law of Thermodynamics.

2.4 Heat, Internal Energy, and Work

- **Internal Energy (U):** The sum of the molecular kinetic energies (translational, rotational, and vibrational) and molecular potential energies in the frame of reference relative to which the centre of mass of the system is at rest. It includes only the energy associated with random, disordered molecular motion, *never* the ordered kinetic energy of the bulk container moving as a whole.
  - U is a macroscopic **thermodynamic state variable**: its value depends solely on the current state of the system, not on the path or history taken to reach that state.
  - For an **ideal gas**, intermolecular forces are neglected (zero molecular potential energy), so its internal energy U is purely the sum of random molecular kinetic energies and depends **only on temperature** T.
- **Heat (Q):** Energy in transit between a system and its surroundings solely due to a temperature difference between them.
- **Work (W):** Energy transfer brought about by macroscopic mechanical means that do not involve a temperature difference (such as moving a piston attached to a weight).

Crucial Conceptual Distinction: Heat (Q) and work (W) are **not** state variables; they are modes of energy transfer (path functions). Saying "a gas in a given state contains a certain amount of heat" or "contains a certain amount of work" is scientifically meaningless. One can only state that a gas in a given state has a certain *internal energy* U, or that a certain amount of heat ΔQ was supplied or work ΔW was done during a process.



Tutorial Diagram 3: (a, b) Internal energy includes only random internal molecular energy in the centre-of-mass rest frame. (c) Heat and work are two distinct modes of changing the state and internal energy of a gas.

2.5 First Law of Thermodynamics

The **First Law of Thermodynamics** is the general law of conservation of energy applied to a system exchanging heat and work with its surroundings:

$$\Delta Q = \Delta U + \Delta W \quad \Longleftrightarrow \quad \Delta Q - \Delta W = \Delta U$$

Standard Sign Conventions:

- $\Delta Q > 0$: Heat is **supplied to (added to)** the system by the surroundings.
- $\Delta Q < 0$: Heat is **released from (removed from)** the system to the surroundings.
- $\Delta W > 0$: Work is done **by the system** on the surroundings (expansion, $\Delta V > 0$).
- $\Delta W < 0$: Work is done **on the system** by the surroundings (compression, $\Delta V < 0$).
- $\Delta U > 0$: Internal energy of the system increases; $\Delta U < 0$: Internal energy decreases.

Path Independence vs. Path Dependence: When a gas goes from initial state (P_1, V_1) to final state (P_2, V_2) via different paths, ΔQ and ΔW individually depend on the path taken, whereas ΔU (and therefore the combination $\Delta Q - \Delta W$) depends *only* on the initial and final states. For a gas expanding against a constant pressure P by a volume change ΔV:

$$\Delta W = P\Delta V \quad \Longrightarrow \quad \Delta Q = \Delta U + P\Delta V$$

2.6 Specific Heat Capacity, Solids, Water, and Mayer's Relation

When heat ΔQ supplied to a substance changes its temperature from T to $T + \Delta T$:

- **Heat Capacity (S):** $S = \dfrac{\Delta Q}{\Delta T}$ (SI unit: $\mathrm{J\,K^{-1}}$, proportional to mass).
- **Specific Heat Capacity (s):** Heat capacity per unit mass m:
$$s = \frac{S}{m} = \frac{1}{m}\frac{\Delta Q}{\Delta T} \quad (\text{SI unit: } \mathrm{J\,kg^{-1}\,K^{-1}})$$
- **Molar Specific Heat Capacity (C):** Heat capacity per mole μ:
$$C = \frac{S}{\mu} = \frac{1}{\mu}\frac{\Delta Q}{\Delta T} \quad (\text{SI unit: } \mathrm{J\,mol^{-1}\,K^{-1}})$$

Molar Specific Heat of Solids (Law of Equipartition of Energy): Consider a solid of N atoms vibrating in three dimensions about their mean positions. A 1D oscillator has average energy $2 \times \frac{1}{2}k_BT = k_BT$, so in 3D each atom has average energy $3k_BT$. For 1 mole (N_A atoms), the internal energy is $U = 3k_BTN_A = 3RT$. Since volume change ΔV for a solid at constant pressure is negligible ($\Delta Q = \Delta U + P\Delta V \approx \Delta U$):

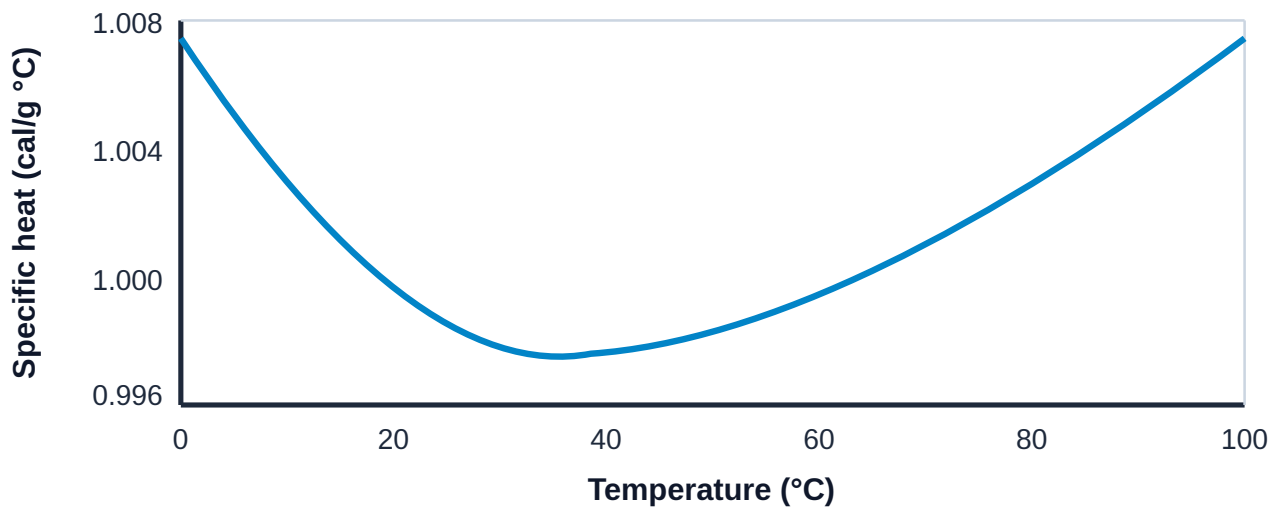
$$C = \frac{\Delta Q}{\Delta T} \approx \frac{\Delta U}{\Delta T} = 3R \approx 24.9\,\mathrm{J\,mol^{-1}\,K^{-1}}$$

This theoretical prediction agrees closely with room-temperature experimental measurements for most solids, with carbon being a notable exception (and the agreement breaks down at low temperatures).

Table 1: Specific and Molar Heat Capacities of Selected Solids at Room Temperature and Atmospheric Pressure

Substance	Specific Heat Capacity s ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	Molar Specific Heat Capacity C ($\mathrm{J\,mol^{-1}\,K^{-1}}$)
Aluminium	900.0	24.4
Carbon	506.5	6.1
Copper	386.4	24.5
Lead	127.7	26.5
Silver	236.1	25.5
Tungsten	134.4	24.9

Specific Heat Capacity of Water and the Calorie: Because the specific heat capacity of water varies slightly between $0\,^\circ\mathrm{C}$ and $100\,^\circ\mathrm{C}$, **one calorie** ($1\,\mathrm{cal}$) is precisely defined as the amount of heat required to raise the temperature of $1\,\mathrm{g}$ of water from $14.5\,^\circ\mathrm{C}$ to $15.5\,^\circ\mathrm{C}$. In SI units, the specific heat capacity of water is $4186\,\mathrm{J\,kg^{-1}\,K^{-1}} = 4.186\,\mathrm{J\,g^{-1}\,K^{-1}}$ (so $1\,\mathrm{cal} = 4.186\,\mathrm{J}$).



Tutorial Diagram 4: Slight variation of the specific heat capacity of water with temperature from 0 °C to 100 °C.

Derivation of Mayer's Relation ($C_p - C_v = R$) for an Ideal Gas: For 1 mole of an ideal gas, $\Delta Q = \Delta U + P\Delta V$.

1. At constant volume ($\Delta V = 0$): $C_v = \left(\frac{\Delta Q}{\Delta T}\right)_v = \left(\frac{\Delta U}{\Delta T}\right)_v = \frac{\Delta U}{\Delta T}$ (subscript dropped because U of an ideal gas depends only on T).

2. At constant pressure: $C_p = \left(\frac{\Delta Q}{\Delta T}\right)_p = \left(\frac{\Delta U}{\Delta T}\right)_p + P\left(\frac{\Delta V}{\Delta T}\right)_p = C_v + P\left(\frac{\Delta V}{\Delta T}\right)_p$.

3. Differentiating the 1-mole ideal gas equation $PV = RT$ at constant pressure P gives $P\left(\frac{\Delta V}{\Delta T}\right)_p = R$. Substituting this yields:

$$C_p - C_v = R$$

2.7 Thermodynamic State Variables and Equation of State

State variables describe **only equilibrium states**. A gas expanding freely into a vacuum after removing a partition, or a fuel-air mixture undergoing an explosive chemical reaction, passes through non-equilibrium states where pressure and temperature

are not uniform throughout the bulk; such intermediate states cannot be plotted as points on a thermodynamic P-V-T surface.

The mathematical relationship connecting state variables for an equilibrium state is the **equation of state**. For μ moles of an ideal gas:

$$PV = \mu RT$$

For a fixed μ, only two variables are independent (e.g., P and V, or T and V). A pressure-volume curve at fixed temperature is called an **isotherm**.

Extensive vs. Intensive Variables: Imagine dividing a system in equilibrium into two equal halves:

- **Extensive Variables:** Depend on the size/mass of the system and get *halved* in each part. Examples: Internal energy (U), Volume (V), Total mass (M), number of moles (μ), and heat/work increments (ΔQ, ΔW).
- **Intensive Variables:** Independent of system size and remain *unchanged* in each part. Examples: Pressure (P), Temperature (T), and Density (ρ).
- *Consistency Rule:* In any valid thermodynamic equation such as $\Delta Q = \Delta U + P\Delta V$, every term must have the same character (here, the product of intensive P and extensive ΔV is extensive, matching ΔQ and ΔU).

2.8 Thermodynamic Processes

Quasi-Static Process: An idealized, infinitely slow process in which the system remains in thermal and mechanical equilibrium with its surroundings at every intermediate stage. Throughout a quasi-static process, the external pressure $P + \Delta P$ and reservoir temperature $T + \Delta T$ differ only *infinitesimally* from the system's P and T.

Table 2: Summary of Special Thermodynamic Processes for an Ideal Gas

Process Type	Defining Condition	Equation / Constraint	Work Done (W) by Gas	First Law Consequence
Isothermal	$T = \text{constant}$	$PV = \text{constant}$ (Boyle's Law); $\Delta U = 0$	$W = \mu RT \ln\left(\frac{V_2}{V_1}\right)$	$Q = W$ (Heat supplied is entirely converted to work)
Adiabatic	$\Delta Q = 0$ (Insulated wall)	$PV^{\gamma} = \text{constant}$; $TV^{\gamma-1} = \text{constant}$ ($\gamma = C_p/C_v$)	$W = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{\mu R(T_1 - T_2)}{\gamma - 1}$	$\Delta U = -W$ (Expansion $W > 0 \Rightarrow T_2 < T_1$; Compression $W < 0 \Rightarrow T_2 > T_1$)
Isochoric	$V = \text{constant}$ ($\Delta V = 0$)	$\frac{P}{T} = \text{constant}$	$W = 0$	$\Delta Q = \Delta U = \mu C_v \Delta T$
Isobaric	$P = \text{constant}$	$\frac{V}{T} = \text{constant}$	$W = P(V_2 - V_1) = \mu R(T_2 - T_1)$	$\Delta Q = \Delta U + W = \mu C_p \Delta T$
Cyclic	Final state = Initial state	$\Delta U_{\text{cycle}} = 0$	Net area enclosed by loop on P-V diagram	$Q_{\text{net}} = W_{\text{net}}$

2.9 Second Law of Thermodynamics, Reversibility, and the Carnot Engine

While the First Law requires energy conservation, many energy-conserving processes never occur spontaneously (e.g., a book on a table never jumps upward by spontaneously cooling the table; heat never flows spontaneously from a colder body to a hotter body). The **Second Law of Thermodynamics** establishes the direction of natural processes and sets fundamental limits on heat engines and refrigerators through two equivalent statements:

> - **Kelvin-Planck Statement:** "No process is possible whose sole result is the absorption of heat from a reservoir and the complete conversion of the heat into work." (Implies heat engine efficiency η can never be 1 or 100%.)
> - **Clausius Statement:** "No process is possible whose sole result is the transfer of heat from a colder object to a hotter object." (Implies the coefficient of performance α of a refrigerator can never be infinite.)

Reversible and Irreversible Processes: A process ($i \to f$) is **reversible** if it can be turned back such that both the system and the surroundings return to their original states with no other change anywhere else in the universe. A process is reversible *if and only if*:

1. It is **quasi-static** (no finite temperature or pressure differences at any stage), and
2. There are **no dissipative effects** (no friction, viscosity, turbulence, etc.).

Spontaneous natural processes—such as free expansion of a gas into vacuum, diffusion of leaking cooking gas, combustion, heat conduction across a finite temperature difference, or stirring a liquid—are **irreversible**.

The Carnot Engine and Carnot Cycle: Proposed by Sadi Carnot in 1824, a **Carnot engine** is a reversible heat engine operating between two temperatures: a hot reservoir (source) at T_1 and a cold reservoir (sink) at T_2. Because heat transfer must occur without a finite temperature difference (isothermally) and temperature changes between T_1 and T_2 must occur without heat exchange (adiabatically), the Carnot cycle consists of four reversible steps:

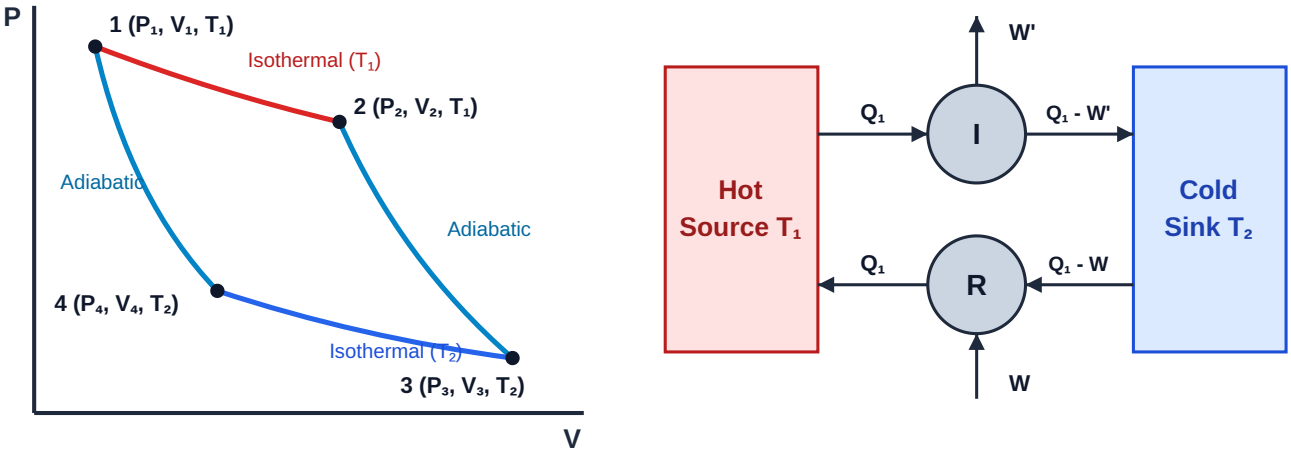
1. **Step $1 \to 2$ (Isothermal Expansion at T_1):** State changes from (P_1, V_1, T_1) to (P_2, V_2, T_1). Heat absorbed from source equals work done by gas:
$$W_{1\to2} = Q_1 = \mu R T_1 \ln\left(\frac{V_2}{V_1}\right)$$
2. **Step $2 \to 3$ (Adiabatic Expansion from T_1 to T_2):** State changes from (P_2, V_2, T_1) to (P_3, V_3, T_2). Work done by gas:
$$W_{2\to3} = \frac{\mu R(T_1 - T_2)}{\gamma - 1}$$
3. **Step $3 \to 4$ (Isothermal Compression at T_2):** State changes from (P_3, V_3, T_2) to (P_4, V_4, T_2). Heat Q_2 released to sink equals work done on the gas:
$$W_{3\to4} = Q_2 = \mu R T_2 \ln\left(\frac{V_3}{V_4}\right)$$
4. **Step $4 \to 1$ (Adiabatic Compression from T_2 to T_1):** State changes from (P_4, V_4, T_2) to (P_1, V_1, T_1). Work done on the gas:
$$W_{4\to1} = \frac{\mu R(T_1 - T_2)}{\gamma - 1}$$

Since $W_{2\to3} = W_{4\to1}$, the net work done by the gas in one cycle is $W = W_{1\to2} - W_{3\to4} = Q_1 - Q_2$. Applying the adiabatic relation $TV^{\gamma-1} = \text{constant}$ to steps $2 \to 3$ and $4 \to 1$:

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \quad \text{and} \quad T_1 V_1^{\gamma-1} = T_2 V_4^{\gamma-1} \quad \Longrightarrow \quad \frac{V_2}{V_1} = \frac{V_3}{V_4}$$

Substituting $\ln(V_3/V_4) = \ln(V_2/V_1)$ into the efficiency formula $\eta = \dfrac{W}{Q_1} = 1 - \dfrac{Q_2}{Q_1}$ gives:

$$\eta = 1 - \frac{Q_2}{Q_1} = 1 - \frac{T_2}{T_1} \quad \text{and} \quad \frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$



(a) Carnot Cycle on P-V Diagram **(b) Proof of Carnot's Theorem**

Tutorial Diagram 5: (a) Carnot cycle on a P-V diagram. (b) Coupling an irreversible engine I with a reversible Carnot refrigerator R to prove Carnot's Theorem.

Carnot's Theorem and Proof:
(a) *Working between two given temperatures T_1 and T_2, no engine can have efficiency greater than that of the Carnot engine.*
(b) *The efficiency of the Carnot engine is independent of the nature of the working substance.*
Proof: Couple an irreversible engine I (absorbing Q_1 at T_1, doing work W', rejecting $Q_1 - W'$ at T_2) with a reversible Carnot engine R run backward as a refrigerator (returning Q_1 to T_1, absorbing $Q_2 = Q_1 - W$ from T_2, requiring work input

W). If $\eta_I > \eta_R$, then $W' > W$. The composite I-R system leaves the hot source T_1 unchanged while extracting net heat $(Q_1 - W) - (Q_1 - W') = W' - W$ from the cold sink T_2 and converting it *entirely* into net work $W' - W$, violating the Kelvin-Planck statement. Thus $\eta_I \leq \eta_R$.

3. Important Formulas, Dimensions & Physical Quantities

Table 3: Physical Quantities, Dimensions, Units, and Remarks

Quantity	Symbol	Dimensions	SI Unit	Key Relation / Remark
Coefficient of volume expansion	α_V	$[\mathrm{K}^{-1}]$	K^{-1}	$\alpha_V = 3\alpha_l$ (for isotropic solids)
Heat supplied to a system	ΔQ	$[\mathrm{M\,L^2\,T^{-2}}]$	J	Q is not a state variable (path-dependent)
Specific heat capacity	s	$[\mathrm{L^2\,T^{-2}\,K^{-1}}]$	$\mathrm{J\,kg^{-1}\,K^{-1}}$	$s = \dfrac{1}{m}\dfrac{\Delta Q}{\Delta T}$
Molar specific heat capacity	C	$[\mathrm{M\,L^2\,T^{-2}\,mol^{-1}\,K^{-1}}]$	$\mathrm{J\,mol^{-1}\,K^{-1}}$	$C = \dfrac{1}{\mu}\dfrac{\Delta Q}{\Delta T}$; $C_p - C_v = R$
Thermal conductivity	K	$[\mathrm{M\,L\,T^{-3}\,K^{-1}}]$	$\mathrm{J\,s^{-1}\,m^{-1}\,K^{-1}}$	Heat current $H = -KA\dfrac{\mathrm{d}T}{\mathrm{d}x}$

4. Problem-Solving Methods & Conceptual Connections

- **Comparing Isothermal and Adiabatic Slopes on a P-V Diagram:** For an isotherm ($PV = \text{constant}$), $\dfrac{\mathrm{d}P}{\mathrm{d}V} = -\dfrac{P}{V}$. For an adiabat ($PV^\gamma = \text{constant}$), $\dfrac{\mathrm{d}P}{\mathrm{d}V} = -\gamma\dfrac{P}{V}$. Since $\gamma = \dfrac{C_p}{C_v} > 1$, an adiabatic curve is always **steeper** than an isothermal curve at their point of intersection by a factor of γ.
- **Work from P-V Graphs:** The work done by a system in any quasi-static process from V_1 to V_2 equals the signed area under the P-V curve projected onto the volume axis ($W = \int_{V_1}^{V_2} P\,\mathrm{d}V$). In a closed cycle, a clockwise loop yields $W_{\text{net}} > 0$ and a counter-clockwise loop yields $W_{\text{net}} < 0$.
- **Free Expansion Shortcut:** When an insulated gas expands into an evacuated chamber (free expansion), $\Delta Q = 0$ (insulated) and $W = 0$ (no external opposing pressure in vacuum). By the First Law, $\Delta U = 0$. For an ideal gas, $\Delta U = 0 \implies \Delta T = 0$.

5. Complete Worked Examples

Worked Example 1: Internal Energy Change During Phase Transition (Boiling of Water)

Problem: Calculate the change in internal energy when $1\ \mathrm{g}$ of water at atmospheric pressure ($P = 1.013 \times 10^5\ \mathrm{Pa}$) transitions from its liquid phase to its vapour phase. The measured latent heat of vaporization of water is $2256\ \mathrm{J\,g^{-1}}$. At atmospheric pressure, $1\ \mathrm{g}$ of water has a volume of $1\ \mathrm{cm^3}$ in the liquid phase and $1671\ \mathrm{cm^3}$ in the vapour phase.

Solution:

1. Heat supplied to $1\ \mathrm{g}$ of water during vaporization at constant pressure and temperature:
$$\Delta Q = mL = 1\ \mathrm{g} \times 2256\ \mathrm{J\,g^{-1}} = 2256\ \mathrm{J}$$
2. Work done by the expanding water-vapour system against constant atmospheric pressure P:
$$\Delta W = P(V_g - V_l) = 1.013 \times 10^5\ \mathrm{Pa} \times (1671 - 1) \times 10^{-6}\ \mathrm{m^3}$$
$$\text{Using textbook approximation } V_g - V_l \approx 1671 \times 10^{-6}\ \mathrm{m^3}: \quad \Delta W = 1.013 \times 10^5 \times 1671 \times 10^{-6} = 169.2\ \mathrm{J}$$
(Note: Using $1670\ \mathrm{cm^3}$ gives $169.17\ \mathrm{J} \approx 169.2\ \mathrm{J}$).
3. Applying the First Law of Thermodynamics ($\Delta Q = \Delta U + \Delta W$):
$$\Delta U = \Delta Q - \Delta W = 2256\ \mathrm{J} - 169.2\ \mathrm{J} = 2086.8\ \mathrm{J}$$

Conclusion: Most of the latent heat ($2086.8\ \mathrm{J}$ out of $2256\ \mathrm{J}$) goes into increasing the internal molecular potential energy of water during the liquid-to-vapour transition, while $169.2\ \mathrm{J}$ performs expansion work against the atmosphere.

Worked Example 2: Fuel Consumption Rate in a Water Geyser

Problem: A geyser heats water flowing at the rate of $3.0\,\text{litres}$ per minute from $27\,^\circ\text{C}$ to $77\,^\circ\text{C}$. If the geyser operates on a gas burner, what is the rate of consumption of the fuel if its heat of combustion is $4.0 \times 10^4\,\text{J g}^{-1}$? (Take specific heat capacity of water $s = 4.2 \times 10^3\,\text{J kg}^{-1}\,\text{K}^{-1}$ or $4186\,\text{J kg}^{-1}\,\text{K}^{-1}$, density of water $= 1.0\,\text{kg L}^{-1}$).

Solution:

Mass of water flowing per minute: $m = 3.0\,\text{L min}^{-1} \times 1.0\,\text{kg L}^{-1} = 3.0\,\text{kg min}^{-1} = 3000\,\text{g min}^{-1}$.
Temperature rise: $\Delta T = 77\,^\circ\text{C} - 27\,^\circ\text{C} = 50\,^\circ\text{C} = 50\,\text{K}$.
Rate of heat required by the water ($\Delta Q/\Delta t$):

$$\frac{\Delta Q}{\Delta t} = ms\Delta T = (3.0\,\text{kg min}^{-1})(4.2 \times 10^3\,\text{J kg}^{-1}\,\text{K}^{-1})(50\,\text{K}) = 6.3 \times 10^5\,\text{J min}^{-1}$$

Let r (in g min^{-1}) be the rate of fuel consumption. Given heat of combustion $L_c = 4.0 \times 10^4\,\text{J g}^{-1}$:

$$r = \frac{\Delta Q/\Delta t}{L_c} = \frac{6.3 \times 10^5\,\text{J min}^{-1}}{4.0 \times 10^4\,\text{J g}^{-1}} = 15.75\,\text{g min}^{-1} \quad (\text{or } 15.70\,\text{g min}^{-1} \text{ using } s = 4186\,\text{J kg}^{-1}\,\text{K}^{-1})$$

Worked Example 3: Isobaric Heating of a Diatomic Gas (Nitrogen)

Problem: What amount of heat must be supplied to $2.0 \times 10^{-2}\,\text{kg}$ of nitrogen at room temperature to raise its temperature by $45\,^\circ\text{C}$ at constant pressure? (Molecular mass of $\text{N}_2 = 28\,\text{g mol}^{-1}$; $R = 8.3\,\text{J mol}^{-1}\,\text{K}^{-1}$).

Solution:

Mass of nitrogen $m = 2.0 \times 10^{-2}\,\text{kg} = 20\,\text{g}$. Number of moles $\mu = \dfrac{m}{M} = \dfrac{20}{28} = \dfrac{5}{7}\,\text{mol}$.

For a diatomic gas like N_2 at room temperature, molar heat capacity at constant pressure is $C_p = \dfrac{7}{2}R$.

Heat supplied at constant pressure:

$$\Delta Q = \mu C_p \Delta T = \left(\frac{5}{7}\,\text{mol}\right)\left(\frac{7}{2} \times 8.3\,\text{J mol}^{-1}\,\text{K}^{-1}\right)(45\,\text{K}) = 2.5 \times 8.3 \times 45 = 933.75\,\text{J} \approx 934\,\text{J}$$

Worked Example 4: Qualitative Conceptual Analysis of Thermal Phenomena

Problem: Explain physically why:
(a) Two bodies at different temperatures T_1 and T_2 brought into thermal contact do not necessarily settle to the mean temperature $(T_1 + T_2)/2$.
(b) The coolant in a chemical or nuclear plant should have a high specific heat capacity.
(c) Air pressure in a car tyre increases during driving.
(d) The climate of a harbour town is more temperate than that of a town in a desert at the same latitude.

Solution:

(a) By conservation of thermal energy in an insulated contact, heat lost by the hotter body equals heat gained by the colder body: $m_1 s_1 (T_1 - T_f) = m_2 s_2 (T_f - T_2)$. The final equilibrium temperature is $T_f = \dfrac{S_1 T_1 + S_2 T_2}{S_1 + S_2}$, which equals the arithmetic mean $\dfrac{T_1 + T_2}{2}$ *only if* the two bodies have equal thermal heat capacities ($S_1 = m_1 s_1 = m_2 s_2 = S_2$).

(b) Since $\Delta Q = ms\Delta T$, a liquid with a high specific heat capacity s can absorb a large quantity of heat ΔQ per unit mass for a small permissible rise in temperature ΔT, preventing plant components from overheating.

(c) During driving, flexing of the tyre and friction with the road perform mechanical work that increases the internal energy and temperature T of the air trapped inside the nearly fixed volume V of the tyre. Since $P \propto T$ at nearly constant volume, the air pressure rises.

(d) Water has a much higher specific heat capacity than desert sand. For the same solar heat input during the day (or radiative heat loss at night), the temperature of the sea water near a harbour changes much less than that of desert sand, moderating the coastal climate through land and sea breezes.

Worked Example 5: Adiabatic Compression of Hydrogen

Problem: A cylinder with a movable piston contains $3\,\text{moles}$ of hydrogen at standard temperature and pressure. The walls of the cylinder are made of a heat insulator, and the piston is insulated by having a pile of sand on it. By what factor does the pressure of the gas increase if the gas is compressed to half its original volume?

Solution:

Because the cylinder and piston are thermally insulated, the compression is **adiabatic** ($\Delta Q = 0$). Hydrogen (H_2) is a diatomic gas, for which $\gamma = \dfrac{C_p}{C_v} = \dfrac{7}{5} = 1.4$.

Using the adiabatic relation $P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$ with $V_2 = \dfrac{V_1}{2}$:

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma} = 2^{1.4} \approx 2.64$$

Thus, the pressure increases by a factor of $2^{1.4} \approx 2.64$.

Worked Example 6: Path Independence of ΔU Between Two Equilibrium States

Problem: In changing the state of a gas adiabatically from an equilibrium state A to another equilibrium state B, an amount of work equal to $22.3\ \text{J}$ is done on the system. If the gas is taken from state A to B via a process in which the net heat absorbed by the system is $9.35\ \text{cal}$, how much is the net work done by the system in the latter case? (Take $1\ \text{cal} = 4.19\ \text{J}$).

Solution:

Process 1 (Adiabatic, $A \to B$): $\Delta Q_1 = 0$, and work is done *on* the system, so $\Delta W_1 = -22.3\ \text{J}$.

$$\Delta U = U_B - U_A = \Delta Q_1 - \Delta W_1 = 0 - (-22.3\ \text{J}) = +22.3\ \text{J}$$

Process 2 (Non-adiabatic, $A \to B$): Since internal energy U is a state variable, $\Delta U = +22.3\ \text{J}$ is identical for any path from A to B.

Heat absorbed: $\Delta Q_2 = 9.35\ \text{cal} \times 4.19\ \text{J cal}^{-1} = 39.1765\ \text{J} \approx 39.2\ \text{J}$.

Net work done by the system in the second process:

$$\Delta W_2 = \Delta Q_2 - \Delta U = 39.1765\ \text{J} - 22.3\ \text{J} = 16.8765\ \text{J} \approx 16.9\ \text{J}$$

Worked Example 7: Free Expansion of an Ideal Gas Between Connected Cylinders

Problem: Two cylinders A and B of equal capacity are connected to each other via a stopcock. Cylinder A contains an ideal gas at standard temperature and pressure ($P_i = 1\ \text{atm}$), while B is completely evacuated. The entire system is thermally insulated. The stopcock is suddenly opened. Determine:

(a) The final pressure of the gas in A and B.

(b) The change in internal energy of the gas.

(c) The change in temperature of the gas.

(d) Whether the intermediate states of the system (before settling to the final equilibrium state) lie on its P-V-T surface.

Solution:

(a) Since the system is thermally insulated ($\Delta Q = 0$) and expands into a vacuum ($\Delta W = 0$), this is an adiabatic free expansion. By the First Law, $\Delta U = 0$, so the final temperature equals the initial temperature ($T_f = T_i$). Because the volume doubles ($V_f = 2V_i$) at constant temperature, Boyle's law $P_f V_f = P_i V_i$ gives $P_f = \dfrac{P_i}{2} = 0.5\ \text{atm}$.

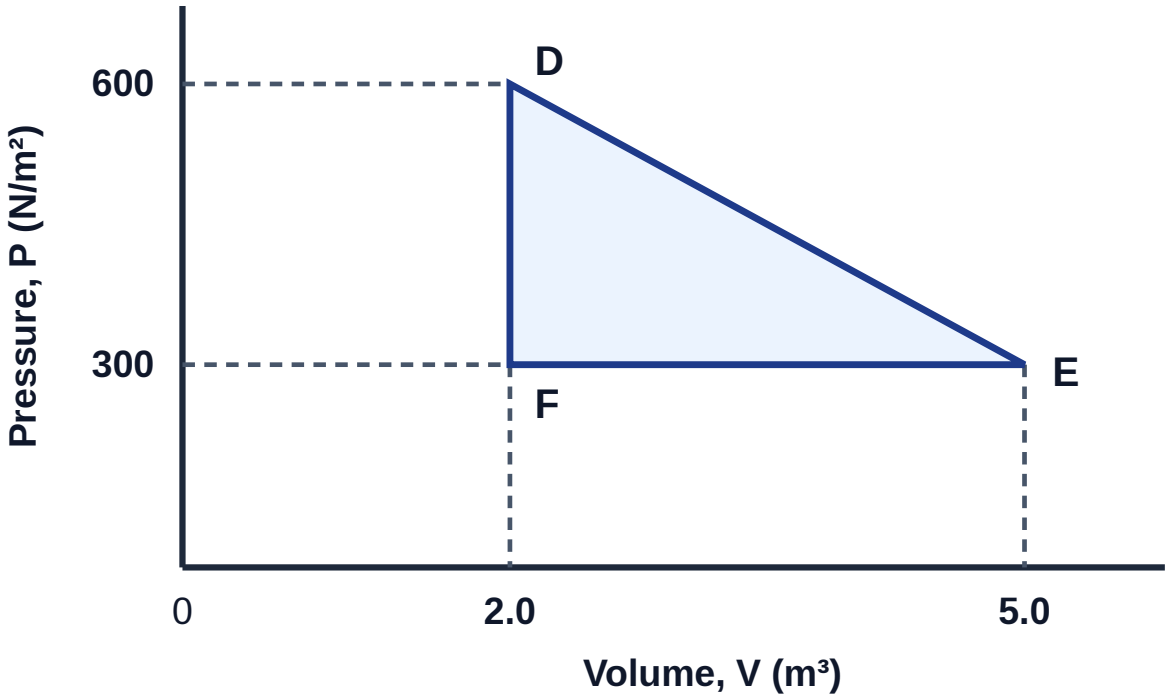
(b) $\Delta U = \Delta Q - \Delta W = 0 - 0 = 0$.

(c) Since U of an ideal gas depends only on temperature, $\Delta U = 0 \implies \Delta T = 0$.

(d) No. Free expansion is a rapid, non-equilibrium, non-quasi-static process in which pressure and temperature are not well-defined throughout the gas until the final equilibrium state is reached. Therefore, intermediate states do not lie on the equilibrium P-V-T surface.

Worked Example 8: Rate Form of the First Law and Indicator Diagram Area

Part I: An electric heater supplies heat to a system at a rate of $100\,\mathrm{W}$. If the system performs work at a rate of $75\,\mathrm{J\,s^{-1}}$, at what rate is the internal energy increasing?
Part II: A thermodynamic system is taken from an original state D to an intermediate state E by the linear process shown in Tutorial Diagram 6 below. Its volume is then reduced to the original value from E to F by an isobaric process. Calculate the total work done by the gas from D to E to F.



Tutorial Diagram 6: Linear expansion D → E followed by isobaric compression E → F (and isochoric return F → D).

Solution:

Part I: Differentiating the First Law with respect to time t:

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \frac{\mathrm{d}Q}{\mathrm{d}t} - \frac{\mathrm{d}W}{\mathrm{d}t} = 100\,\mathrm{W} - 75\,\mathrm{W} = 25\,\mathrm{W} = 25\,\mathrm{J\,s^{-1}}$$

Part II:

• Work done during expansion $D \to E$ (area of trapezoid under DE):

$$W_{DE} = +\frac{1}{2}(P_D + P_E)(V_E - V_D) = \frac{1}{2}(600 + 300)\,\mathrm{N\,m^{-2}} \times (5.0 - 2.0)\,\mathrm{m^3} = 450 \times 3.0 = +1350\,\mathrm{J}$$

• Work done during isobaric compression $E \to F$:

$$W_{EF} = P_E(V_F - V_E) = 300\,\mathrm{N\,m^{-2}} \times (2.0 - 5.0)\,\mathrm{m^3} = -900\,\mathrm{J}$$

• Total work done from $D \to E \to F$ (which also equals the area of right triangle $\triangle DEF$ since $W_{FD} = 0$):

$$W_{DEF} = W_{DE} + W_{EF} = 1350\,\mathrm{J} - 900\,\mathrm{J} = \frac{1}{2}(600 - 300)(5.0 - 2.0) = +450\,\mathrm{J}$$

6. Exam-Style Practice Questions (NEET UG Standard)

Note: All questions below are original practice items designed strictly in accordance with the NEET UG syllabus and examination pattern (4 choices, single correct answer).

Practice Question 1 (Conceptual — State vs. Path Variables)

An ideal gas is taken from an initial equilibrium state (P_1, V_1) to a final equilibrium state (P_2, V_2) along three distinct quasi-static paths I, II, and III. Let ΔQ_i, ΔW_i, and ΔU_i ($i = \mathrm{I}, \mathrm{II}, \mathrm{III}$) denote the heat supplied, work done by the gas, and change in internal energy along each path. Which of the following relations is always true?

(A) $\Delta Q_{\mathrm{I}} = \Delta Q_{\mathrm{II}} = \Delta Q_{\mathrm{III}}$

(B) $\Delta W_{\mathrm{I}} = \Delta W_{\mathrm{II}} = \Delta W_{\mathrm{III}}$

(C) $\Delta Q_{\mathrm{I}} - \Delta W_{\mathrm{I}} = \Delta Q_{\mathrm{II}} - \Delta W_{\mathrm{II}} = \Delta Q_{\mathrm{III}} - \Delta W_{\mathrm{III}}$

(D) $\Delta Q_{\mathrm{I}} + \Delta W_{\mathrm{I}} = \Delta Q_{\mathrm{II}} + \Delta W_{\mathrm{II}} = \Delta Q_{\mathrm{III}} + \Delta W_{\mathrm{III}}$

Correct Answer: (C)

Worked Solution: Heat ΔQ and work ΔW are path-dependent quantities, whereas internal energy U is a thermodynamic state variable. Therefore, $\Delta U = U_f - U_i$ depends only on the initial and final states and is identical for all three paths: $\Delta U_{\mathrm{I}} = \Delta U_{\mathrm{II}} = \Delta U_{\mathrm{III}}$. By the First Law of Thermodynamics ($\Delta U = \Delta Q - \Delta W$), the combination $\Delta Q - \Delta W$ is path-independent.

Practice Question 2 (Statement-Based — Extensive and Intensive Variables)

Consider the following two statements regarding thermodynamic variables:
Statement I: Internal energy U, volume V, and total mass M are extensive variables, whereas pressure P, temperature T, and density ρ are intensive variables.
Statement II: In the First Law relation $\Delta Q = \Delta U + P\Delta V$, the term $P\Delta V$ is an intensive quantity because pressure P is intensive.
In the light of the above statements, choose the correct option:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Answer: (C)

Worked Solution: Statement I is correct: dividing a system in equilibrium into two equal parts halves U, V, and M (extensive), while leaving P, T, and ρ unchanged (intensive). Statement II is incorrect because the product of an intensive variable (P) and an extensive variable (ΔV) is an *extensive* quantity, ensuring dimensional and thermodynamic consistency with ΔQ and ΔU.

Practice Question 3 (Numerical — Work in Isothermal Expansion)

Two moles of an ideal gas undergo a quasi-static isothermal expansion at $300\,\mathrm{K}$ from an initial volume of $2.0\,\mathrm{L}$ to a final volume of $20.0\,\mathrm{L}$. How much heat is absorbed by the gas during this process? (Given: $R = 8.31\,\mathrm{J\,mol^{-1}\,K^{-1}}$ and $\ln 10 = 2.303$)

(A) $5.74\,\mathrm{kJ}$

(B) $11.48\,\mathrm{kJ}$

(C) $2.49\,\mathrm{kJ}$

(D) Zero

Correct Answer: (B)

Worked Solution: In an isothermal process of an ideal gas, $\Delta T = 0 \implies \Delta U = 0$. By the First Law, heat absorbed equals work done by the gas:

$$Q = W = \mu RT \ln\left(\frac{V_2}{V_1}\right) = 2 \times 8.31 \times 300 \times \ln\left(\frac{20.0}{2.0}\right) = 4986 \times 2.303 = 11482.76\,\mathrm{J} \approx 11.48\,\mathrm{kJ}$$

Practice Question 4 (Numerical — Fraction of Heat Converted to Work in Isobaric Process)

An ideal diatomic gas at room temperature ($\gamma = \frac{7}{5}$) expands isobarically when an amount of heat ΔQ is supplied to it. The ratio of the work done by the gas to the heat supplied ($\Delta W / \Delta Q$) is:

(A) $\dfrac{2}{5}$

(B) $\dfrac{2}{7}$

(C) $\dfrac{5}{7}$

(D) $\dfrac{3}{5}$

Correct Answer: (B)

Worked Solution: For an isobaric process of μ moles of an ideal gas:

$$\Delta W = P\Delta V = \mu R \Delta T \quad \text{and} \quad \Delta Q = \mu C_p \Delta T$$

Using $C_p - C_v = R$ and $\gamma = \dfrac{C_p}{C_v} = \dfrac{7}{5}$, we have $C_p = \dfrac{\gamma R}{\gamma - 1} = \dfrac{7}{2}R$. Therefore:

$$\frac{\Delta W}{\Delta Q} = \frac{\mu R \Delta T}{\mu C_p \Delta T} = \frac{R}{C_p} = 1 - \frac{1}{\gamma} = \frac{2}{7}$$

Practice Question 5 (Numerical — Work Done in Adiabatic Process)

Half a mole ($\mu = 0.5\,\text{mol}$) of an ideal monatomic gas ($\gamma = \frac{5}{3}$) expands adiabatically and quasi-statically such that its temperature drops from $400\,\text{K}$ to $280\,\text{K}$. The work done by the gas is: (Take $R = 8.3\,\text{J mol}^{-1}\,\text{K}^{-1}$)

(A) $498\,\text{J}$

(B) $747\,\text{J}$

(C) $996\,\text{J}$

(D) $-747\,\text{J}$

Correct Answer: (B)

Worked Solution: For an adiabatic process of an ideal gas from $T_1 = 400\,\text{K}$ to $T_2 = 280\,\text{K}$:

$$W = \frac{\mu R(T_1 - T_2)}{\gamma - 1} = \frac{0.5 \times 8.3 \times (400 - 280)}{\frac{5}{3} - 1} = \frac{0.5 \times 8.3 \times 120}{\frac{2}{3}} = 0.5 \times 8.3 \times 180 = 747\,\text{J}$$

Practice Question 6 (Assertion–Reason — Isothermal Heat Transfer)

Assertion (A): In a quasi-static isothermal expansion of an ideal gas, heat is absorbed by the gas from the surrounding reservoir even though the temperature of the gas remains equal to that of the reservoir throughout the process.
Reason (R): The heat transfer is made possible by an infinitesimal temperature difference between the surrounding reservoir and the gas at every stage of the quasi-static process.
Choose the correct answer from the options given below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Answer: (A)

Worked Solution: In an ideal quasi-static isothermal process, the temperature of the surrounding reservoir ($T + \Delta T$ with $\Delta T \to 0$) differs only infinitesimally from the system temperature T. This infinitesimal difference drives heat transfer into the gas to compensate for the work done by the gas ($Q = W$) without producing a finite temperature change.

Practice Question 7 (Numerical — Carnot Engine Efficiency Changes)

A Carnot engine operating between a hot reservoir at temperature T_1 and a cold sink at temperature T_2 has an efficiency of $\frac{1}{4}$. When the temperature of the cold sink is reduced by $65\,\text{K}$ (keeping T_1 unchanged), its efficiency increases to $\frac{1}{2}$. The temperatures T_1 and T_2 are respectively:

(A) $260\,\text{K}$ and $195\,\text{K}$

(B) $390\,\text{K}$ and $260\,\text{K}$

(C) $300\,\text{K}$ and $225\,\text{K}$

(D) $520\,\text{K}$ and $390\,\text{K}$

Correct Answer: (A)

Worked Solution: For a Carnot engine, $\eta = 1 - \frac{T_2}{T_1}$.

Initially: $\eta_1 = 1 - \frac{T_2}{T_1} = \frac{1}{4} \implies \frac{T_2}{T_1} = \frac{3}{4} \implies T_2 = 0.75T_1$.

After lowering the sink temperature by $65\,\text{K}$:

$$\eta_2 = 1 - \frac{T_2 - 65}{T_1} = \frac{1}{2} \implies \frac{T_2}{T_1} - \frac{65}{T_1} = \frac{1}{2}$$

Substituting $\frac{T_2}{T_1} = \frac{3}{4}$:

$$\frac{3}{4} - \frac{65}{T_1} = \frac{1}{2} \implies \frac{65}{T_1} = \frac{1}{4} \implies T_1 = 260\,\text{K}$$

Then $T_2 = \frac{3}{4} \times 260\,\text{K} = 195\,\text{K}$.

Practice Question 8 (Numerical — Carnot Cycle Volume Compression Ratios)

An ideal monatomic gas ($\gamma = \frac{5}{3}$) undergoes a Carnot cycle between a source at $T_1 = 600\,\text{K}$ and a sink at $T_2 = 150\,\text{K}$. If the volume doubles during the isothermal expansion step ($V_2 = 2V_1$), then the ratio of the maximum volume to the minimum volume in the entire cycle (V_3/V_1) is:

(A) 8

(B) 16

(C) 32

(D) 4

Correct Answer: (B)

Worked Solution: In the Carnot cycle:

• Step $1 \to 2$ (isothermal expansion at T_1): $\frac{V_2}{V_1} = 2$.

• Step $2 \to 3$ (adiabatic expansion from $T_1 = 600\,\text{K}$ to $T_2 = 150\,\text{K}$):

$$T_1 V_2^{\gamma-1} = T_2 V_3^{\gamma-1} \implies \frac{V_3}{V_2} = \left(\frac{T_1}{T_2}\right)^{\frac{1}{\gamma-1}} = \left(\frac{600}{150}\right)^{\frac{1}{5/3-1}} = 4^{3/2} = 8$$

Therefore, the overall ratio of maximum volume V_3 to minimum volume V_1 is:

$$\frac{V_3}{V_1} = \left(\frac{V_3}{V_2}\right)\left(\frac{V_2}{V_1}\right) = 8 \times 2 = 16$$

Practice Question 9 (Matching Type — Thermodynamic Processes and Features)

Match List-I (Thermodynamic Process/System of an Ideal Gas) with List-II (Characteristic Relation) and select the correct option:

List-I	List-II
(P) Molar heat capacity of a solid at room temperature (Equipartition law)	(1) $\Delta U = 0$ and $\Delta Q = \Delta W \neq 0$
(Q) Quasi-static isothermal expansion of an ideal gas	(2) $\Delta Q = 0, \Delta W = 0, \Delta U = 0$
(R) Adiabatic free expansion of an ideal gas into vacuum	(3) $C = 3R$
(S) Reversible Carnot cycle between T_1 and T_2	(4) $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

(A) P–3, Q–1, R–2, S–4

(B) P–3, Q–2, R–1, S–4

(C) P–4, Q–1, R–2, S–3

(D) P–1, Q–3, R–2, S–4

Correct Answer: (A)

Worked Solution:
• (P) For a 3D solid of N_A oscillating atoms, $U = 3RT \implies C = 3R$ (matches 3).
• (Q) In quasi-static isothermal expansion of an ideal gas, T is constant ($\Delta U = 0$) and $\Delta Q = \Delta W = \mu RT \ln(V_2/V_1) \neq 0$ (matches 1).
• (R) In adiabatic free expansion into vacuum, no heat enters ($\Delta Q = 0$) and no work is done ($\Delta W = 0$), so $\Delta U = 0$ (matches 2).
• (S) For a Carnot cycle, the universal thermodynamic temperature relation gives $Q_1/Q_2 = T_1/T_2$ (matches 4).

Practice Question 10 (Conceptual — Second Law and Carnot's Theorem)

Two reversible Carnot engines E_A and E_B operate between the same hot reservoir at $500\,\mathrm{K}$ and the same cold sink at $300\,\mathrm{K}$. Engine E_A uses $1\,\mathrm{mole}$ of helium (monatomic, $\gamma = \frac{5}{3}$) as its working substance, while engine E_B uses $2\,\mathrm{moles}$ of oxygen (diatomic, $\gamma = \frac{7}{5}$). If η_A and η_B are their respective thermal efficiencies, then:

(A) $\eta_A > \eta_B$ because $\gamma_{\mathrm{He}} > \gamma_{\mathrm{O_2}}$

(B) $\eta_A < \eta_B$ because E_B uses twice the number of moles

(C) $\eta_A = \eta_B = 40\%$

(D) $\eta_A = \eta_B = 60\%$

Correct Answer: (C)

Worked Solution: By Carnot's Theorem, the efficiency of any reversible Carnot engine operating between two given temperatures T_1 and T_2 is completely independent of the nature or amount of the working substance. Thus:

$$\eta_A = \eta_B = 1 - \frac{T_2}{T_1} = 1 - \frac{300}{500} = 1 - 0.60 = 0.40 = 40\%$$

7. Common Pitfalls & Final Revision Checklist

Common Exam Traps to Avoid:

- **Confusing $\Delta U = 0$ with $\Delta Q = 0$:** In an *isothermal* process of an ideal gas, $\Delta T = 0 \implies \Delta U = 0$, so $\Delta Q = \Delta W \neq 0$. In an *adiabatic* process, $\Delta Q = 0 \implies \Delta U = -\Delta W \neq 0$ (unless it is a free expansion where $\Delta W = 0$ as well).
- **Including Bulk Motion in Internal Energy:** Never add $\frac{1}{2}MV_{\text{cm}}^2$ of a moving gas container or a flying bullet to its thermodynamic internal energy U.
- **Using Celsius in Carnot Efficiency:** Always convert T_1 and T_2 to the absolute Kelvin scale ($T(\text{K}) = t(^\circ\text{C}) + 273.15$) before computing $\eta = 1 - \dfrac{T_2}{T_1}$.
- **Sign of Work in Physics vs. Chemistry:** In physics, ΔW in $\Delta Q = \Delta U + \Delta W$ is the work done *by* the system ($+P\Delta V$). Always check whether work is done *by* ($+W$) or *on* ($-W$) the gas.

Concise Final Recap Checklist:

1. **Zeroth Law:** $T_A = T_C$ and $T_B = T_C \implies T_A = T_B$ (defines temperature).
2. **First Law:** $\Delta Q = \Delta U + \Delta W$ (conservation of energy; U is a state variable, Q and W are path functions).
3. **Specific Heats:** $s = \dfrac{1}{m}\dfrac{\Delta Q}{\Delta T}$, $C = \dfrac{1}{\mu}\dfrac{\Delta Q}{\Delta T}$; for solids $C \approx 3R$; for ideal gases $C_p - C_v = R$ and $\gamma = \dfrac{C_p}{C_v}$.
4. **Work Formulas:** Isothermal $W = \mu RT \ln\left(\dfrac{V_2}{V_1}\right)$; Adiabatic $W = \dfrac{\mu R(T_1 - T_2)}{\gamma - 1}$; Isobaric $W = P\Delta V = \mu R \Delta T$; Isochoric $W = 0$.
5. **Second Law & Carnot Engine:** Prohibits 100% efficient heat engines (Kelvin-Planck) and self-acting heat transfer from cold to hot (Clausius). Maximum reversible efficiency is $\eta_{\text{Carnot}} = 1 - \dfrac{T_2}{T_1} = 1 - \dfrac{Q_2}{Q_1}$.